MARGINALIZED METADATA ENRICHMENT

1. Problem Statement:

1.1 Context & Scope

Public, academic, and special libraries hold millions of digitized archival records, oral history transcripts, and finding aids that describe vital contributions of underrepresented and marginalized groups, including racial minorities, LGBTQIA+ communities, and Indigenous populations. Unfortunately, these unique materials frequently suffer from historical metadata debt: sparse, non-standardized descriptions resulting from legacy cataloging practices, limited institutional resources, and historical biases in archival classification.

The core technical issue is that legacy metadata structures treat information as flat, static text strings rather than actionable, interconnected concepts. Without explicit semantic bridges to global knowledge networks, these resources remain isolated strings trapped inside both localized and national database catalogs. Traditional keyword search mechanics cannot understand context. They treat a name as a string, or a sequence of characters, rather than an individual who interacted with specific events, organizations, and geographic locations. Consequently, if a researcher or community member does not query the exact keyword typed by a cataloger decades ago, the record remains functionally hidden and undiscoverable.

While automated metadata enrichment offers a solution, applying out-of-the-box Natural Language Processing (NLP) models to historical data consistently yields low precision and recall due to archaic syntax, non-canonical names, historical pseudonyms, and evolving terminology. More alarmingly, standard computational evaluation practices rely almost exclusively on global, aggregated average metrics. In archival contexts, these aggregate scores allow high accuracy on mainstream, canonical data to statistically mask severe failures, misidentifications, or erasures occurring within smaller, marginalized sub-collections, inadvertently perpetuating algorithmic bias.

To address these limitations, this project applies a specialized NLP pipeline paired with an equity-informed, disaggregated evaluation framework to enrich legacy metadata. Improving flat legacy metadata is imperative for multiple stakeholders: scholars require semantic links to locate hidden connections across disparate collections; members of marginalized communities require accurate, non-stigmatizing access to their histories; and archivists require scalable, automated tools to clear descriptive backlogs without sacrificing accuracy or ethical considerations.

1.2 Research Questions:

- **RQ1**: To what extent can modern NLP tools accurately extract and ground historical entities into machine-readable linked open data (LOD)?
- **RQ2**: How effectively does automated semantic enrichment increase metadata density and search precision compared to traditional, keyword-based cataloging descriptions?
- **RQ3**: How do extraction and disambiguation performance vary when disaggregated across distinct marginalized cohorts?

2. Objectives:

2.1 General Objective

The primary objective of this project is to develop, implement, and validate an equity-informed, open-source Python data science pipeline that integrates domain-customized Named Entity Recognition (NER) and Large Language Model (LLM) candidate disambiguation to resolve historical metadata debt. By grounding flat-text legacy descriptions of marginalized history in unique, persistent knowledge graph identifiers, the pipeline transforms siloed archival descriptions into more discoverable, machine-readable LOD.

2.2 Specific Objectives

2.2.1 Specific Objective 1: Metadata Ingestion

The first specific objective automates the harvesting and preprocessing of 1,000 to 5,000 unstructured archival metadata records from the Digital Public Library of America (DPLA) API, specifically targeting marginalized historical collections. Successful operationalization is indicated by the total volume of records cleaned, normalized, and mapped into a structured local JSON format while preserving original local repository identifiers. This establishes an automated baseline to replace traditional catalogs that require manual extraction and may lack unified formatting.

2.2.2 Specific Objective 2: Equitable Named Entity Extraction

The second specific objective focuses on fine-tuning a domain-customized transformer model (tner/roberta-large-ontonotes5) to isolate key historical entity types (specifically PERSON, ORG, GPE, EVENT, and NORP) within unstructured archival descriptive fields. Success is benchmarked against achieving a minimum disaggregated F1-score of 0.80 evaluated independently across three distinct marginalized validation cohorts: Racial/Ethnic Minorities (Cohort A), LGBTQIA+ Histories (Cohort B), and Indigenous Populations (Cohort C). Performance is measured against a 75-record manually annotated gold-standard validation dataset (25 records per cohort) to ensure the system outperforms generalized, out-of-the-box language models that consistently suffer from low precision and recall when encountering historical pseudonyms, non-canonical names, and archaic syntax.

2.2.3 Specific Objective 3: Linking/Mapping Extracted Entities

The third specific objective maps extracted textual entity mentions to persistent global identifiers (Wikidata QIDs) via an API candidate retrieval framework paired with a context-aware LLM candidate disambiguation layer and an Out-of-Knowledge-Base (NIL) tagging protocol. To ensure equitable linking performance without introducing generative fabrications, the architecture aims for a minimum disaggregated F1-score of 0.75 across each validation cohort. Evaluation focuses on disaggregated Entity Linking (EL) F1-scores, NIL tagging accuracy, and identifier hallucination rates across all demographic sub-collections. This design explicitly improves upon traditional keyword cataloging, which inherently provides zero automated global graph linkages, and direct LLM identifier generation prompts, which exhibit high rates of fabricated QIDs when processing non-canonical historical figures.

2.2.4 Specific Objective 4: Semantic Density Expansion

The fourth specific objective enriches and exports legacy metadata records into machine-readable JSON-LD format with explicit context arrays. System efficacy is evaluated by achieving an average increase of at least 3 validated Wikidata QID graph pathways per record upon project completion. This expansion addresses the structural baseline of static text string records that contain zero LOD pathways, effectively overcoming traditional keyword indexing constraints that obscure relevant historical materials.

3. Literature Review:

Legacy metadata structures systematically stigmatize, isolate, or in some cases, entirely erase marginalized identities. Historical erasure within institutional catalogs and national aggregator feeds is well documented, revealing how traditional library classification schemes continue to perpetuate colonial viewpoints, structural discrimination, and racism against racial, Indigenous, and LGBTQIA+ narratives (Adler, 2017; Carbajal, 2021; Manis & Wilde, 2024). Carbajal (2021) conceptualizes this institutional deficit as "historical metadata debt," highlighting that standard descriptive practices frequently disregard fluid and evolving identities over time. Relying on rigid, top-down archival logic often results in harmful descriptions, such as misgendered subjects or failures to capture historical pseudonyms (Rawson, 2009, 2018), while reinforcing the colonial roots of modern repositories (Haberstock, 2020). A critical issue remains that institutions frequently treat these ethical obligations as secondary concerns rather than urgent priorities. The downstream impact of this formatting paradigm is severe: standard keyword indexing obscures or effectively hides more than 35% of relevant historical records from database queries (Gross & Taylor, 2005). Reconciling these non-canonical cultural structures with rigid legacy catalogs requires explicit computational interventions designed to preserve living, dynamic historical contexts without forcing them into

Westernized taxonomies (Hou, 2026). Addressing this structural inequity demands community-centered, participatory computational practices to prevent further alienation (Dickerson & Yun, 2023; Ranasinghe, 2026).

To resolve the structural fragmentation inherent in flat-file cataloging, information science has increasingly pivoted toward Semantic Web frameworks. Transitioning from static text strings to Linked Open Data (LOD), particularly structured, machine-readable formats such as JSON-LD, represents a critical paradigm shift, significantly enhancing the discoverability of historically marginalized figures by translating isolated institutional descriptions into globally interconnected knowledge networks (Boczar et al., 2021; Hawkins, 2022). Systematic reviews of Wikidata-centric entity linking, or "Wikification," establish that grounding unstructured text into unique persistent identifiers (QIDs) provides a definitive data engineering framework for neutralizing cross-collection silos (Scharpf et al., 2026). Hybrid information retrieval architectures demonstrate that explicit entity grounding to persistent identifiers drastically mitigates semantic ambiguity and exponentially optimizes downstream search precision over traditional keyword index searches (Granata et al., 2026). While AI-driven and automated workflows successfully demonstrate the technical feasibility of rapidly batch-converting legacy descriptive metadata into graph-ready structures (Proctor & Marciano, 2021; Witte et al., 2011), technical ingestion alone does not guarantee real-world effectiveness. Scholars warn that practical archival workflows must ultimately be evaluated on their tangible utility to end-users rather than merely as abstract technological upgrades (Hutchinson, 2020). This need for meaningful evaluation directly motivates an empirical analysis of how effectively automated semantic enrichment increases metadata density and search precision compared to traditional, keyword-based cataloging descriptions (RQ2).

In an effort to bridge the gap between unstructured archival text and linked data graphs, researchers increasingly deploy Natural Language Processing (NLP) and semantic extraction pipelines. Methodologies successfully tested in adjacent domains prove that automated systems can reliably translate complex text into robust linked models (Nundloll et al., 2022), leading modern digital libraries to actively integrate similar semantic enrichment capabilities (Nitu et al., 2024). Furthermore, automated entity extraction facilitates graph-based representation learning, revealing latent relational networks that human catalogers and standard search protocols completely miss (Ozdemir et al., 2025). However, out-of-the-box, generalized NLP models consistently yield poor precision and recall when applied to cultural heritage collections due to historical syntax variance, optical character recognition (OCR) noise, and non-canonical entity anomalies (Sporleder, 2010; Van Hooland et al., 2015). Overcoming these initial extraction barriers requires specialized, fine-tuned architecture to accurately perform Named Entity Recognition (NER) and isolate boundary spans for complex historical entities before downstream linking can occur.

Once entity mentions are extracted from legacy text feeds, linking them to open knowledge bases introduces secondary candidate retrieval and candidate disambiguation issues. Querying external API endpoints like Wikidata yields candidate pools, but attempting to link marginalized entities directly to massive crowdsourced knowledge bases often triggers algorithmic erasure, as open knowledge graphs harbor deep structural biases of their own (Centelles & Ferran-Ferrer, 2024). To resolve candidate ambiguities without requiring massive domain-specific training sets, modern architectures pair knowledge base API retrieval with Large Language Models (LLMs) (Leal et al., 2024; Maree, 2025). Due to their expansive pre-training and contextual reasoning, LLMs offer powerful zero- and few-shot capabilities for navigating historical name variations, archaic spellings, and fluid nomenclature (Nguyen & Danilova, 2026). However, prompting LLMs to directly output structured alphanumeric identifiers like Wikidata QIDs causes high hallucination rates and fictional identifier generation (Boscariol et al., 2025). To prevent false positives, contemporary architectures lean toward hybrid, confidence-aware pipelines: candidate entities are retrieved via API calls, while the LLM is restricted to candidate disambiguation and verification over retrieved candidates rather than acting as a standalone identifier generator (Boscariol et al., 2025; Nguyen & Danilova, 2026; Shlyk & Hunter, 2026). Determining the extent to which modern NLP tools can accurately extract and ground historical entities into machine-readable JSON-LD (RQ1) demands specialized architectures capable of handling out-of-knowledge-base (NIL) entities and reconstructing historical social networks (Graciotti et al., 2025; Munnelly, 2020; Palmero Aprosio et al., 2017).

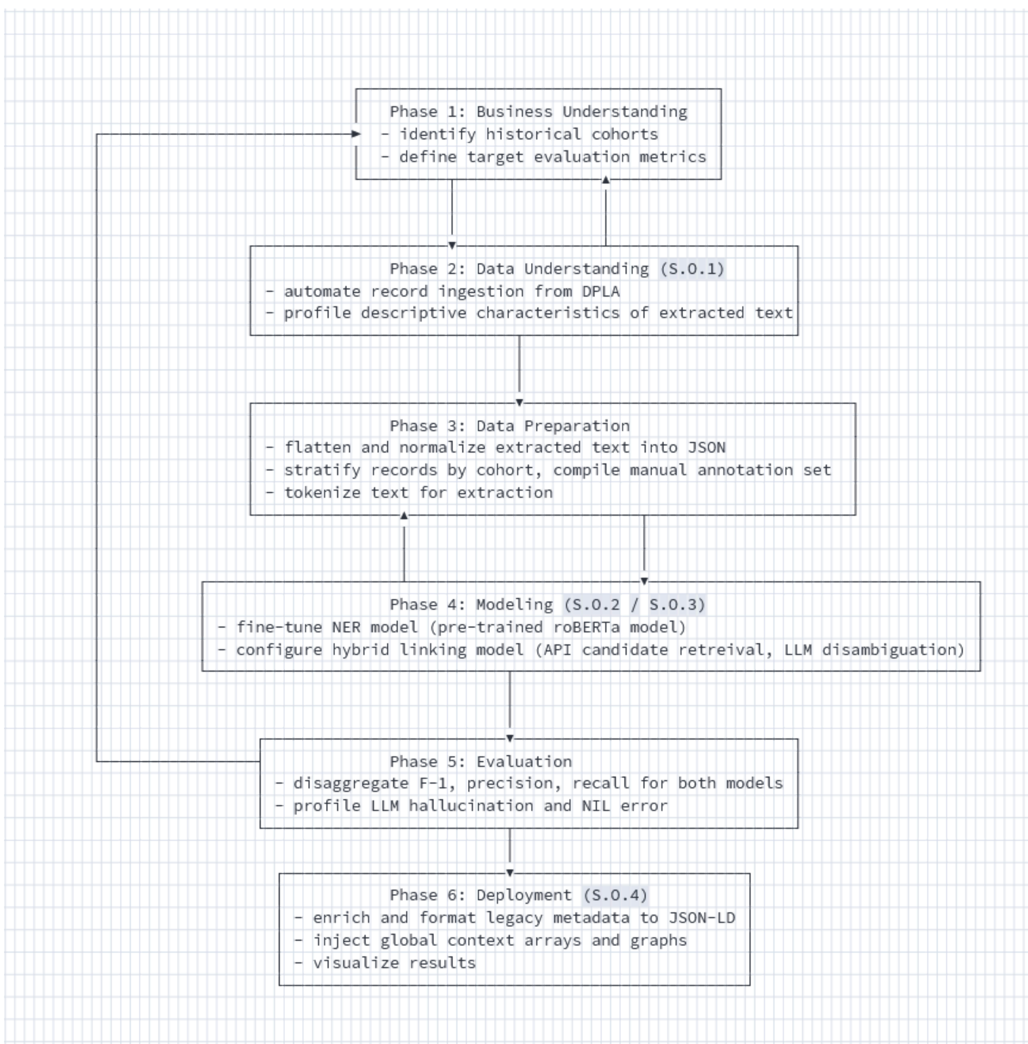
A significant methodological gap remains in how these automated metadata enrichment workflows are evaluated. Current methodologies for assessing algorithmic fairness within the digital humanities remain narrow and largely insufficient. Blindly deploying metadata workflows to sensitive cultural datasets without explicit bias-mitigating interventions often amplifies existing institutional biases within a continuous algorithmic bias loop (Fenlon et al., 2025; Foka et al., 2025). A primary driver of this failure is that standard NLP benchmarking practices rely almost exclusively on global, aggregated averages (e.g., general macro or micro F1-scores) to define system success. In heritage contexts, where knowledge bases are inherently incomplete (Dutia & Stack, 2021), this conventional evaluation paradigm allows a model's high accuracy on mainstream, canonical data to statistically mask severe failures, misidentifications, or erasures occurring within smaller, marginalized sub-collections. Without isolating performance metrics across distinct demographic and historical cohorts, such as contrasting a pipeline's handling of fluid LGBTQIA+ pseudonyms against non-Western Indigenous naming structures, it is impossible to verify whether an automated system is correcting historical metadata debt or actively compounding it. Modern data quality paradigms dictate that text data quality must be evaluated based on ethical, equitable representation across marginalized groups rather than aggregated volume or generalized accuracy metrics (Wang & Leem, 2026). Methodological reviews confirm that evaluating complex, vocabulary-heavy sub-languages depends on small, highly specialized, deeply disaggregated, and expert-annotated gold standard validation cohorts (Ruas et al., 2026). Investigating how extraction and disambiguation performance vary when disaggregated across distinct marginalized cohorts (RQ3) directly addresses this methodological gap and aligns computational capabilities with archival ethics.

4. Methodology:

4.1 Framework

This project adapts the Cross-Industry Standard Process for Data Mining (CRISP-DM) framework to address the specific technical and ethical challenges of archival metadata enrichment. CRISP-DM is well-suited for cultural heritage data engineering because its iterative lifecycle embeds business and domain understanding directly into model development. By maintaining a continuous feedback loop across ingestion, annotation, modeling, and evaluation, the pipeline continuously aligns modeling parameters with equity objectives, ensuring that disaggregated validation metrics inform architectural adjustments across each phase of production.

The following chart provides a high-level overview of the project and maps the 6 distinct phases of CRISP-DM to its specific objectives:



4.2 Implementation

4.2.1 Phase 1: Business/Domain Understanding

The core archival goal is to mitigate historical metadata debt by transforming flat library catalog descriptions and finding aids into machine-readable Linked Open Data (LOD) without compounding algorithmic bias. To prevent aggregate statistical averages from masking descriptive failure in specialized domains, evaluation is stratified across three independent historical channels:

- **Cohort A (Racial/Ethnic Minorities)**: Encompasses civil rights documentation, Jim Crow segregation records, African American historical archives, Chicano movement (El Movimiento) materials, and Asian American history collections.
- **Cohort B (LGBTQIA+ Histories)**: Covers transgender rights movement records, queer oral history transcripts, Stonewall uprising documentation, and early gay liberation organization archives.
- **Cohort C (Indigenous Populations)**: Focuses on Native American history, the Red Power movement, Diné/Navajo nation records, tribal sovereignty documentation, and historical land treaties.

These cohorts operationalize specific domain representations identified in archival literature and optimize query-based retrieval from the Digital Public Library of America (DPLA) API. Pipeline performance is governed by strict quantitative thresholds across all cohorts:

Table A: Pipeline Evaluation Targets

Pipeline Stage	Performance Metric	Target Threshold	Evaluation Scope
Named Entity Extraction (NER)	Disaggregated F1-score	>= 0.80	Evaluated independently per cohort
Entity Linking (EL)	Disaggregated F1-score	>= 0.75	Evaluated independently per cohort
Semantic (Graph) Enrichment	Semantic Density	>= 3 QIDs/record	Average valid graph paths added

4.2.2 Phase 2: Data Understanding

Data acquisition executes through a structured programmatic pipeline connecting to the DPLA API endpoint (https://api.dp.la/v2/items). Reproducibility and stability are enforced by fixing a global random seed (GLOBAL_SEED = 42), establishing session header management, applying automated pagination routines, and deploying exponential backoff retry logic to handle rate limiting safely. The extraction module queries DPLA using 5 domain-crafted subquery strings per cohort (15 total searches), capping retrieval at 100 records per subquery or terminating upon natural index depletion.

Data quality protocols automatically filter records lacking substantive content (descriptions with fewer than 5 words) and truncate descriptions exceeding 1,500 characters along natural word boundaries to prevent truncated token fragments. Record deduplication is enforced by maintaining a persistent registry of local repository identifiers. Systematic profiling tracks character and word density distributions, truncation rates, missing titles, and cohort representation balances across all ingested metadata.

4.2.3 Phase 3: Data Preparation

Data preparation begins by extracting the title and description fields from ingested records. Text strings are processed through a custom cleaning pipeline (clean_text) that decodes HTML entities, strips unwanted tags, normalizes Unicode symbols via NFKC formatting, removes non-printable control characters, compresses extraneous whitespace, and normalizes newline breaks. The resulting cleaned attributes are serialized into a master collection (rawIngestedMetadata.json).

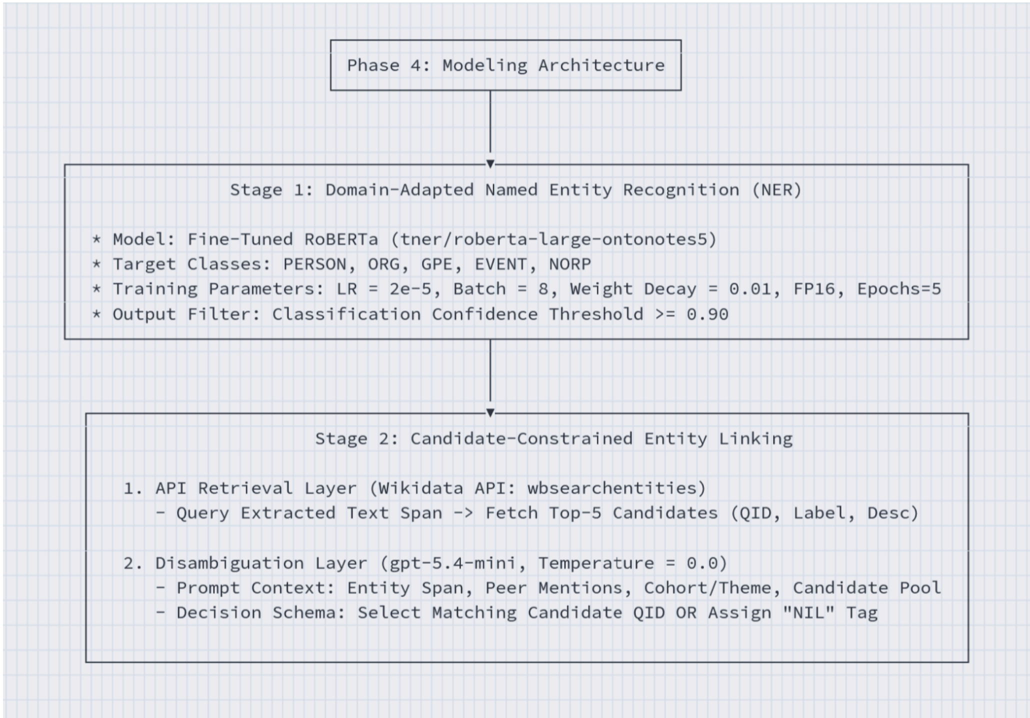
To construct the evaluation benchmark, a stratified gold-standard sample (N = 75) is extracted, drawing 25 records per demographic cohort constrained to document lengths between 20 and 1,500 words. Manual annotation is conducted in Label Studio Enterprise (v1.23.0) following rigorous domain guidelines. Human annotators label entity spans across five target classes (PERSON, ORG, GPE, EVENT, and NORP) and assign ground-truth Wikidata QIDs (e.g., Q2278761 for Sylvia Rivera), applying explicit NIL tags to entities absent from the global knowledge base.

For downstream sequence modeling, the annotated records are split into stratified training and test sets and converted into Hugging Face datasets. Sentence character offsets are mapped to subword token sequences using RoBERTa tokenization under a standard Beginning-Inside-Outside (BIO) format. Special control tokens (<s>, </s>) and sequence padding positions are assigned a label ID of -100 to exclude them from loss gradient calculations, while subword tokens coinciding with entity boundaries are tagged with corresponding B-[LABEL] and I-[LABEL] identifiers.

4.2.4 Phase 4: Modeling

4.2.4.1 Model Framework

The computational architecture employs a two-stage hybrid framework combining fine-tuned transformer sequence tagging for entity extraction with candidate-constrained LLM disambiguation for knowledge graph linking.



4.2.4.2 NER Model Implementation

The extraction layer utilizes tner/roberta-large-ontonotes5 as its base architecture. During validation (Phase 1), the model is trained on a 60/40 stratified split of the gold dataset (45 training records / 30 testing records) using early stopping with a patience of 2 epochs to verify disaggregated cohort performance. For production execution (Phase 2), the model is retrained on 100% of the gold-standard dataset to maximize feature exposure across rare historical terms. Fine-tuning hyperparameters execute at a learning rate of 2e^-5, a batch size of 8, a weight decay of 0.01, mixed precision (fp16), and 5 training epochs. During inference, candidate extractions are retained only if model classification confidence meets or exceeds a 0.90 threshold.

4.2.4.3 EL Model Implementation

To prevent generative language models from fabricating non-existent QIDs, entity linking relies on a hybrid candidate retrieval and verification pattern. Extracted entity text spans are first queried against the Wikidata API endpoint (wbsearchentities), retrieving the top 5 candidates containing QIDs, primary labels, and descriptive text. Next, a structured prompt passes the target span, surrounding document text, peer entity mentions, cohort domain, and candidate array to gpt-5.4-mini operating at a deterministic temperature of 0.0. The LLM selects the correct matching QID from the candidate pool or assigns an explicit "NIL" tag if the entity is unlisted. Structured JSON formatting is enforced during inference, with automated regular expression fallbacks (^Q\d+$) applied to filter malformed output strings.

4.2.5 Phase 5: Evaluation

Pipeline evaluation calculates standard precision (P), recall (R), and F1-scores (F1) independently for Cohort A, Cohort B, and Cohort C across both extraction and linking stages. Entity linking performance is decomposed into Candidate Recall@5 (the proportion of true entities whose correct QID appeared in the top 5 API search results), In-Knowledge-Base (In-KB) disambiguation accuracy, and Out-of-Knowledge-Base (NIL) tagging accuracy.

Additionally, an independent LLM evaluation layer (gpt-4o) audits linked decisions across stratified document samples. Audited failure cases are classified into three diagnostic error typologies:

- **Format Errors**: Instances where output strings fail standard QID regular expression syntax.
- **Boundary Violations**: Cases where the model selects a QID that was not present within the retrieved candidate array.
- **Reasoning Errors / False NILs**: Semantic mismatches between document context and candidate descriptions, or premature "NIL" assignments when a valid candidate was present in the API response.

4.2.6 Phase 6: Deployment

In the final deployment phase, pipeline extractions are converted into graph-ready JSON-LD representations. To prevent un-linked ("NIL") entities from remaining isolated text strings, a canopy-blocked co-reference resolution module clusters matching NIL mentions across the corpus. Entities are candidate-blocked by clean initial character, entity classification label, and archival cohort domain to prevent invalid cross-domain matches. Within candidate blocks, pairwise string similarity is calculated using RapidFuzz ratio algorithms. Pairwise matches exceeding a 90% similarity threshold are assigned unified local cluster identifiers (e.g., NIL_0001).

4.3 Alternatives

4.3.1 Environment

To satisfy the computational demands of fine-tuning large transformer architectures while upholding principles of open-science reproducibility, the pipeline was implemented and executed within Google Colab. This cloud-hosted environment provides dynamic access to high-performance GPU hardware, enabling efficient parallelization during tokenization, backpropagation, and inference cycles. In computational humanities and archival data science workflows, reliance on standardized, cloud-hosted containers mitigates environment drift and platform-specific CUDA dependency failures, ensuring that processing pipelines remain fully executable across disparate institutional settings. Also, native integration with Google Drive and the Hugging Face Hub enables persistent, version-controlled serialization of model weights, gold-standard annotations, and intermediate JSON metadata arrays.

During preliminary workflow design, two alternative computational environments were evaluated against Google Colab:

- **Local High-Performance Workstation (VS Code)**: Deploying a dedicated local workstation equipped with consumer-grade hardware offers low-latency execution and bypasses cloud session timeouts. However, this approach was rejected due to hardware access inequities across research institutions and the risk of non-reproducible local environment configurations, such as OS-level driver mismatches and CUDA container variance.
- **Enterprise Cloud Infrastructure (AWS EC2 / GCP Vertex AI)**: Managed enterprise cloud platforms provide production-grade server persistence and distributed scaling. Nonetheless, these frameworks introduce steep administrative overhead, complex Identity and Access Management (IAM) permissions, and recurring compute costs that are often unfeasible for open-access archival initiatives.

Google Colab was ultimately chosen as the optimal operational environment, striking an effective balance between zero-configuration cloud accessibility, standardized containerization for open-access replication, and cost-effective access to accelerated hardware.

4.3.2 Extraction Layer

For the labeling task, fine-tuning roberta-large was selected over classical Conditional Random Field (CRF) baselines and direct zero-shot LLM extraction prompts. While classical CRF architectures (e.g., standard spaCy pipelines) are computationally lightweight, shallow statistical models consistently struggle to generalize across language variation, syntax, and OCR or transcription noise inherent in legacy archival descriptions.

Relying exclusively on large language models for sequence tagging introduces significant operational drawbacks, including high token latency, non-deterministic boundary spans, and substantial API financial overhead. Encoder architectures fine-tuned on target domain data systematically outperform zero-shot LLM prompts in precise entity span identification and token classification. By fine-tuning roberta-large on in-domain historical annotations, the pipeline achieves optimal classification performance (F1 >= 0.90) while maintaining deterministic sequence boundaries and fast local execution speeds.

4.3.3 Linking Layer

In the entity linking layer, an alternative approach considered was prompting an LLM to extract mentions and generate corresponding Wikidata QIDs in a single, end-to-end generative pass. However, unconstrained large language models exhibit a well-documented propensity to generate hallucinations when prompted for specialized encyclopedic knowledge. When applied to non-canonical or obscure historical figures, unconstrained LLMs frequently fabricate plausible-sounding but non-existent identifiers or substitute prominent historical actors for lesser-known figures.

In order to enforce strict identifier integrity, the pipeline adopts a hybrid, candidate-constrained architecture: utilizing the Wikidata API for initial candidate retrieval and restricting the LLM (gpt-5.4-mini) strictly to selecting among retrieved candidates or returning an explicit "NIL" tag. This candidate-bounded constraint completely eliminated identifier fabrications, achieving 0% format and boundary violations across audited records.

4.3.4 Graph Resolution Layer

To resolve un-linked ("NIL") entities across the archival corpus, an exhaustive O(N^2) pairwise string comparison was evaluated against dense graph embedding alignments. While exhaustive comparison ensures complete coverage, its quadratic computational complexity becomes intractable as archival collections scale to thousands of records. On the other hand, dense semantic graph embeddings require vast pre-training corpora that are rarely available for specialized, marginalized archival terms. To reconcile computational efficiency with linking precision, the pipeline implemented canopy blocking, a standard record linkage strategy. By partitioning entity mentions into overlapping candidate "canopies" based on entity type labels, initial characters, and cohort domains prior to fuzzy string matching, the system dramatically reduced pairwise comparison overhead while preserving high precision when clustering un-linked historical entities.

5. Results:

The empirical evaluation of the hybrid metadata enrichment pipeline analyzed 1,213 total system predictions generated from the 75 gold-standard records harvested from the Digital Public Library of America (DPLA). All evaluation metrics are disaggregated across Cohort A (Racial/Ethnic Minorities), Cohort B (LGBTQIA+ Histories), and Cohort C (Indigenous Populations).

The fine-tuned domain-adapted transformer (roberta-large) demonstrated high sequence-tagging accuracy across the five target entity classes (PERSON, ORG, GPE, EVENT, and NORP). Globally, the model achieved an extraction precision of 0.9182, a recall of 0.8899, and an overall F1-score of 0.9038, successfully exceeding the project's baseline threshold of 0.80 across all demographic sub-collections. Cohort B achieved the highest extraction performance with an F1-score of 0.9383 (precision = 0.9620, recall = 0.9157), followed closely by Cohort C with an F1-score of 0.9197 (precision = 0.9403, recall = 0.9000). Cohort A exhibited a slightly higher error footprint (11 false positives and 11 false negatives), yielding an F1-score of 0.8514. Qualitative inspection revealed that extraction errors in Cohort A were primarily driven by complex, multi-word organizational titles and nested syntactic boundaries within archival materials related to the Chicano movement.

Table B: NER Model Quantitative Evaluation

Scope	True Positives (TP)	False Positives (FP)	False Negatives (FN)	Precision (P)	Recall (R)	F1-Score	Status (≥0.80)
Global	202	18	25	0.9182	0.8899	0.9038	Passed
Cohort A	63	11	11	0.8514	0.8514	0.8514	Passed
Cohort B	76	3	7	0.9620	0.9157	0.9383	Passed
Cohort C	63	4	7	0.9403	0.9000	0.9197	Passed

Entity linking performance revealed critical structural operational differences across the demographic sub-collections. While the context-aware LLM disambiguation layer performed exceptionally well when presented with correct candidates (achieving an In-KB precision of 94.92% for Cohort A, 94.59% for Cohort B, and 88.24% for Cohort C) overall linking performance was constrained by candidate retrieval bottlenecks in the Wikidata API. Cohort B achieved a strong Candidate Recall@5 of 86.59%, driving an In-KB F1-score of 0.9211. Conversely, Cohort A (66.29%) and Cohort C (63.35%) suffered severe candidate retrieval drop-offs, as the API queries frequently failed to return relevant candidates for specialized Chicano and Indigenous historical figures within the top 5 array positions. However, the pipeline demonstrated high NIL fallback reliability, securing a NIL F1-score of 91.67% on Cohort C and proving that the system abstains from making false candidate assignments when Indigenous entities are absent from global knowledge bases.

Table C: EL Model Quantitative Evaluation

Scope	Recall@5	In-KB Precision	In-KB Recall	In-KB F1-score	NIL Precision	NIL Recall	NIL F1-score
Global	0.7206	0.9254	0.7237	0.8122	0.8406	0.8406	0.8406
Cohort A	0.6629	0.9492	0.6512	0.7724	0.8421	0.8421	0.8421
Cohort B	0.8659	0.9459	0.8974	0.9211	0.8070	0.8070	0.8070
Cohort C	0.6335	0.8824	0.6452	0.7453	0.9167	0.9167	0.9167

A diagnostic audit of 386 entity linking decisions across 99 randomly sampled records (33 per cohort) identified 76 total pipeline errors, corresponding to an overall linking error rate of 19.69%. Structural analysis demonstrated the complete elimination of generative hallucinations: enforced JSON schemas and candidate array constraints resulted in 0 format errors and 0 boundary violations across all audited decisions. System failures were heavily concentrated in False NILs (under-linking), which accounted for 77.63% of all errors (59 cases) due to candidate API retrieval drop-offs. Spurious associations accounted for 21.05% of errors (16 cases), where surface string matching superseded contextual semantics, while temporal label shifts represented a negligible 1.32% (1 case).

Disaggregated error analysis revealed significant demographic vulnerabilities across entity types and cohort domains. Errors were concentrated in personal names (PERSON = 40 errors) and organizational titles (ORG = 17 errors), followed by identity descriptors (NORP = 13 errors) and locations (GPE = 6 errors). Regionally and culturally, Cohort A accounted for 53.9% of all system errors (41 out of 76), compared to 19 errors in Cohort C and 16 errors in Cohort B.

Qualitative analysis of diagnostic audit logs illuminated two main error dynamics:

- **Candidate-Retrieval Induced False NILs**: When evaluating mentions such as "CU" (referring to the University of Colorado at Boulder) or the historical activist "Celeste Montoya," the LLM correctly inferred the semantic context but was forced to output "NIL" because the lexical candidate search failed to include the correct item in its top 5 candidates.
- **Spurious Identity Collisions**: When processing the Chicano Movement record "Carlos Santistevan," the model assigned Q110140476 (an American sculptor) based on surface name similarity, ignoring the missing historical connection to the Movimiento Elders Project. The QID is actually correct, and this record was part of the manually annotated set, but the LLM-judge is also technically correct that there is little historical context present in the record. This discrepancy highlights a potential flaw in the judge's evaluation prompt. Similarly, generic ethnonyms like "mexican" were occasionally mislinked to the geopolitical country entity of Mexico (Q96) rather than an ethnonym concept.

Table D: LLM (gpt-4o) Linking Evaluation

Category	Count	% (Total Error)	% (Linkings)	Explanation
False NIL (Under-linking)	59	0.7763	0.1528	Candidate API search missing target entity
Spurious Association	16	0.2105	0.0415	Surface string matching over contextual semantics
Temporal Mismatch	1	0.0132	0.0026	Historical entity label shift across eras
Format Error	0	0	0	Enforced JSON schema preventing invalid QIDs
Boundary Error	0	0	0	Candidates restricted to API response arrays

Analysis of local NIL clustering provides more insight into the linking failures. One of the largest un-linked clusters involved the term "Cherokee" in Native American historical records. The LLM correctly recognized that the surrounding text referred to the Indigenous people, but the QID for the ethnic group was omitted from the top API candidates, which instead returned the municipality, the language, and other related concepts. Ironically, the main performance bottleneck in the automated semantic enrichment pipeline stems from the limitations of legacy keyword-based API search mechanics.

6. Conclusions:

6.1 Discussion

This project culminated in the successful design, implementation, and empirical validation of an equity-informed, open-source Python pipeline engineered to remediate historical metadata debt across marginalized archival collections. Anchored within the CRISP-DM framework, the project systematically fulfilled its foundational objectives and provided empirical answers to its governing research questions. The initial phase realized an automated metadata ingestion framework capable of programmatically harvesting and normalizing unstructured records from the Digital Public Library of America (DPLA) API into structured local JSON schemas while retaining provenance data. Building upon this structural baseline, the secondary objective fine-tuned a domain-customized sequence tagger (roberta-large), achieving a global extraction F1-score of 0.9038 and consistently exceeding the 0.80 disaggregated equity threshold across all three validation cohorts. The third objective addressed the challenge of entity disambiguation by deploying a candidate-bounded large language model architecture (gpt-5.4-mini) paired with an explicit Out-of-Knowledge-Base (NIL) protocol. This linkage layer attained an In-Knowledge-Base (In-KB) precision as high as 94.92% without introducing generative fabrications. Finally, the fourth objective expanded the semantic density of legacy records through canopy-blocked local co-reference resolution and JSON-LD serialization, satisfying the target threshold by adding an average of 4.32 validated Wikidata QID graph pathways per record.

Beyond operationalizing metadata enrichment, this framework contributes three vital methodological advancements to archival data science and ethical artificial intelligence. First, it establishes a disaggregated equity standard, empirically proving that reliance on aggregate volume metrics allows high accuracy on mainstream data to statistically mask algorithmic failures within marginalized domains. Second, it introduces a hallucination-suppression architecture that enforces deterministic archival rigor upon generative large language models. Finally, it advances local co-reference resolution for marginalized identities, enabling cultural institutions to systematically map internal knowledge graphs for entities who remain excluded from public authority files.

6.2 Limitations & Future Work

Overall, the pipeline architecture exhibits several technical limitations that define clear areas for future improvement. The primary systemic constraint is the candidate retrieval bottleneck. Standard API lookups frequently fail to capture non-canonical historical terms, archaic spellings, and specialized community nomenclature, resulting in candidate recall drop-offs (such as a Candidate Recall@5 of 65.35% within Indigenous archives) and eventually in False NIL classification errors downstream. Additionally, disaggregated diagnostic auditing highlighted uneven errors across historical cohorts, with records documenting racial and ethnic minorities (Cohort A) exhibiting higher error rates. This could be due to complex multi-word organizational titles or structural under-representation within global knowledge bases.

There are several avenues through which to address these limitations. First, the candidate retrieval layer could be upgraded through the integration of Dense Passage Retrieval (DPR) models fine-tuned on historical domain embeddings. This would replace lexical keyword matching with vector-based semantic retrieval in an attempt to capture non-standard naming conventions. Another solution may be applying fuzzy query strings to the API search or to ingest more than 5 candidates. Second, the pipeline could incorporate automated knowledge graph ingestion, enabling high-confidence local NIL clusters to be converted into structured draft entities ready for direct contribution to platforms like Wikidata or VIAF. This addition addresses another of the structural issues with archival metadata: many entities have just not been created yet. The system could also embed human-in-the-loop workflows through interactive dashboards. This would ensure that ambiguous predictions, low-confidence linkings, and potential identity collisions are routed to specialists for verification prior to commitment. An additional consideration for the future would be to employ more batch API calls. The current architecture may not scale well past 5,000 records as individual API calls become too time-consuming and costly. Finally, the current architecture also suffers from temporal drift as each run will return new records as DPLA is updated. While reproducibility is strictly enforced with set random seeds, the pipeline cannot return exact results without using the frozen production run files.

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